Revenue

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Revenue

Meaning :

Revenue is the receipt of money from the sale of output by a firm in a given time period.

Concepts of Revenue

- **Total Revenue**
- **Average revenue**
- **Marginal Revenue**

Total Revenue

Total Revenue (TR) is the total amount of money receipts of a firm from the sale of output.

TR = Price X Q where Q is the output sold

TR = AR X Q = $\sum MR$

Average Revenue

Average Revenue (AR) is revenue per unit of output sold.

$$AR = \frac{TR}{output\ Sold}$$

$$AR = \frac{Price\ X\ Output\ Sold}{Output\ Sold}$$

AR=Price

Marginal Revenue

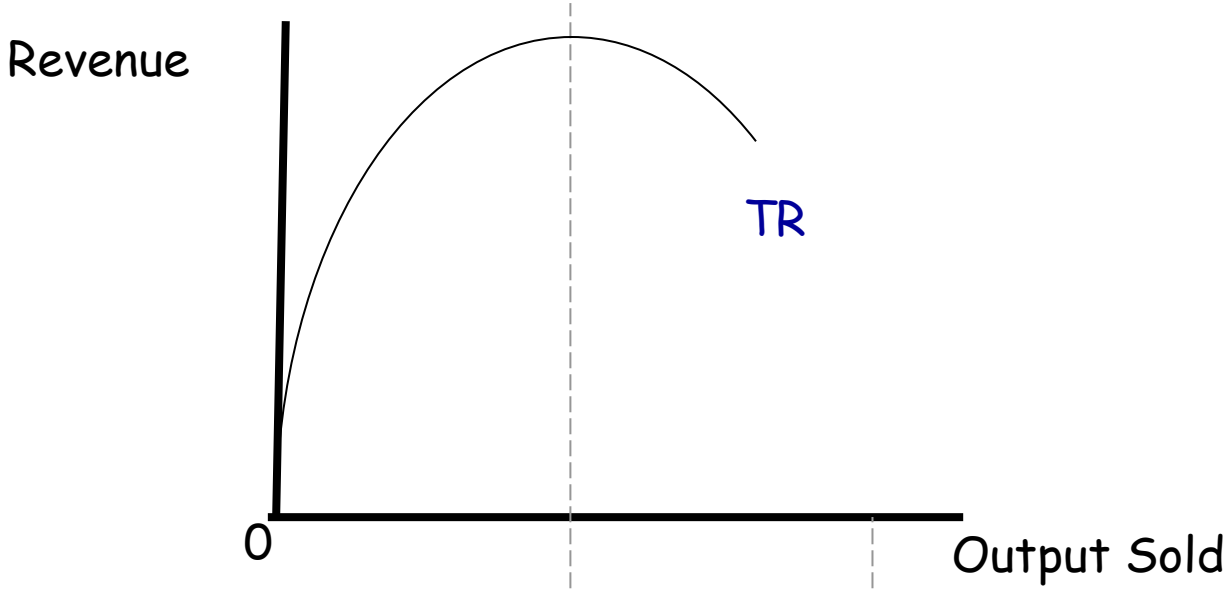
Marginal Revenue (MR) is the rate of change in total revenue with respect to change in output.

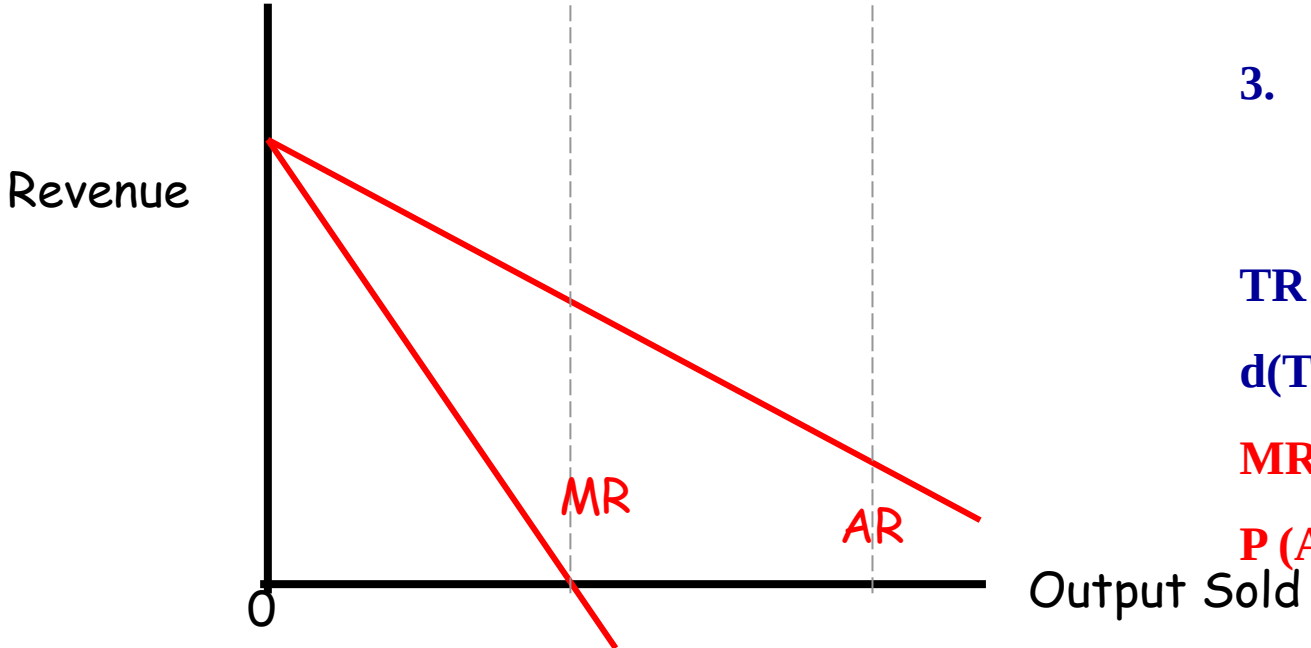
$$MR = \frac{d(TR)}{dQ}$$

Where,

Q is output sold

TR, AR and MR under Im-Perfect Competition





1. **When total Revenue is maximised, MR = 0**

2. **AR curve is the demand curve facing a firm in the market**

3. **AR and MR curves are downward sloping, MR curve lies below AR curve.**

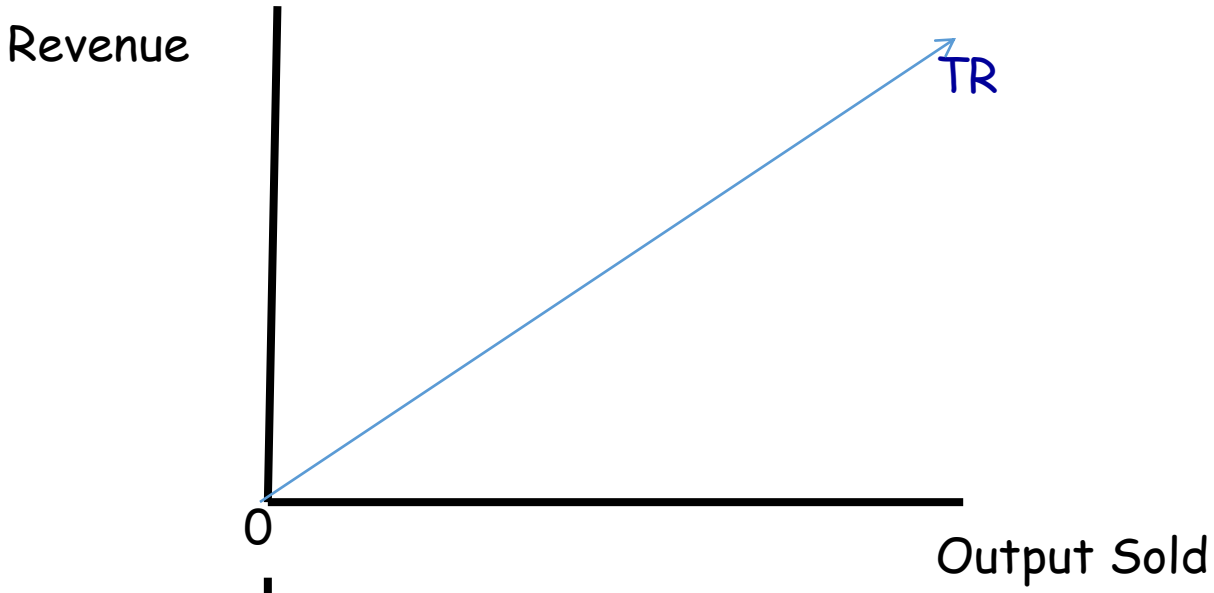
TR = P*Q

d(TR)/dQ= P (dQ/dQ) + Q (dP/dQ)

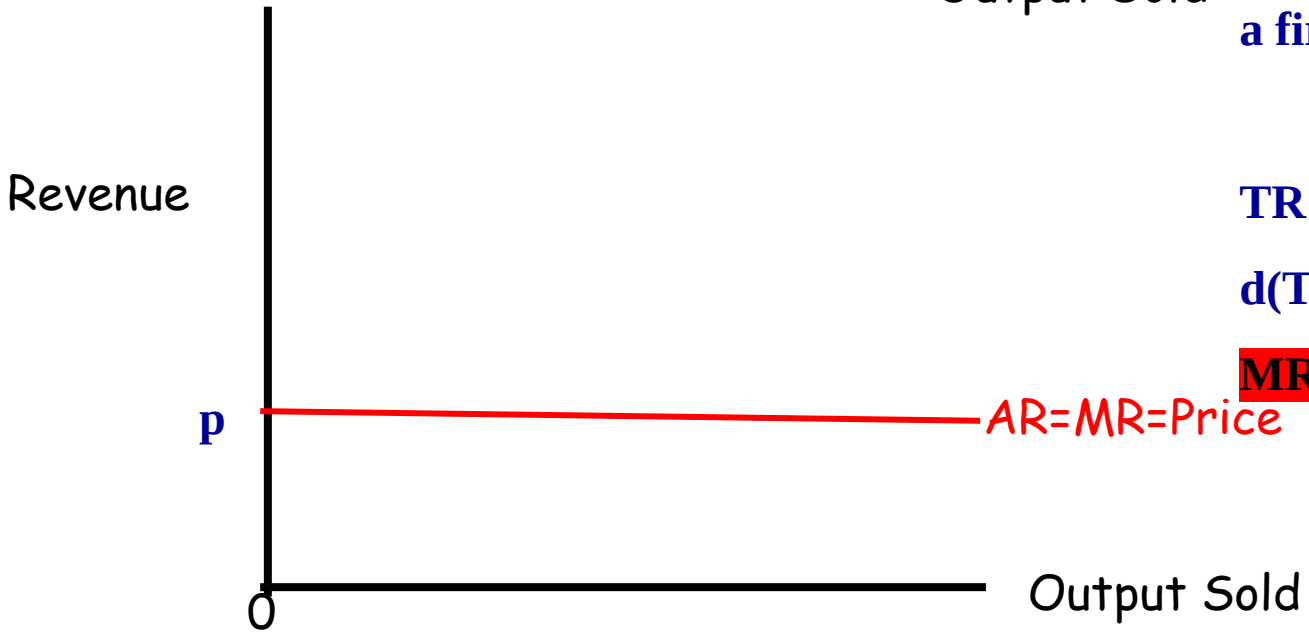
MR= P - Q* dP/dQ

P (AR) = MR+ Q* dP/dQ

TR, AR and MR under Perfect Competition



1. Under perfect Competition price is uniform and given. As such, AR(price) and MR become equal.
2. AR and MR curves coincide and become parallel to output axis.
3. AR curve i.e. the demand curve facing a firm in the market is perfectly elastic.



TR = P*Q

d(TR)/dQ= P (dQ/dQ)

MR = P =AR

MARKET STRUCTURE

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HOMOGENOUS PRODUCTs : IDENTICAL or PERFECT SUBSTITUTES

Example-I

Example-II

DIFFERENTIATED PRODUCTS: SIMILAR BUT NOT IDENTICAL OR DIFFERENT BUT CLOSE SUBSTITUTES

Example-I

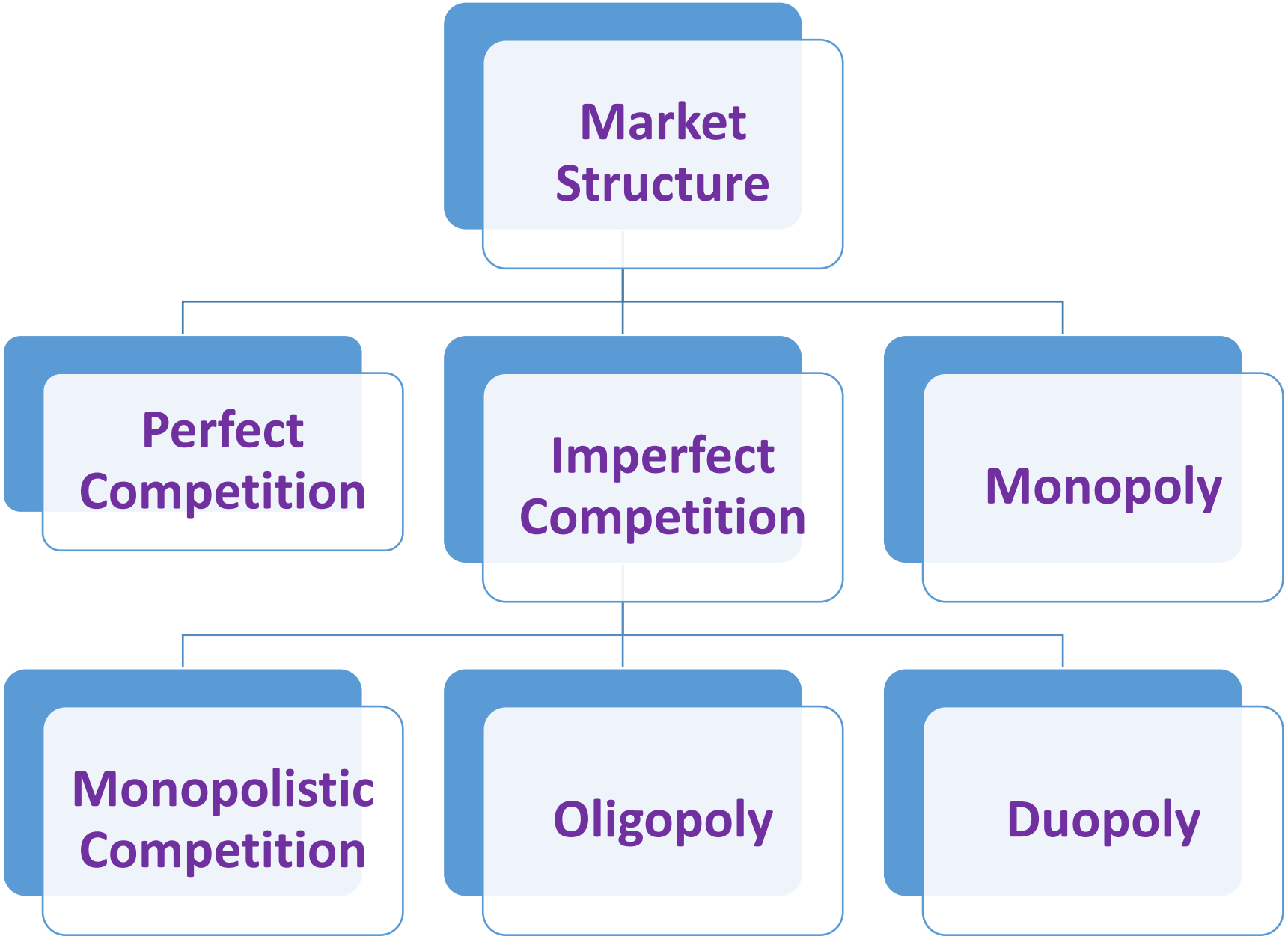
Example-II

MEANING OF MARKET IN ECONOMICS

In common language, market means a place where goods are purchased.

However, in economics, **market means an arrangement which establish effective relationship between buyers and seller of a commodity.**

Hence, each commodity has its own market.


Market Structure
Perfect Competition
Imperfect Competition
Monopoly
Monopolistic Competition
Oligopoly
Duopoly

PERFECT COMPETITION

Perfect competition is a market structure in which large number of sellers sell a homogenous product at uniform price.

PERFECT COMPETITION: FEATURES

1. Large Number of Buyers and Sellers

There are large number of sellers and buyers such that each seller supply a small fraction of the market supply and each buyer buys a small fraction of the market demand. Thus, no single seller or buyer can influence the price and affect the market

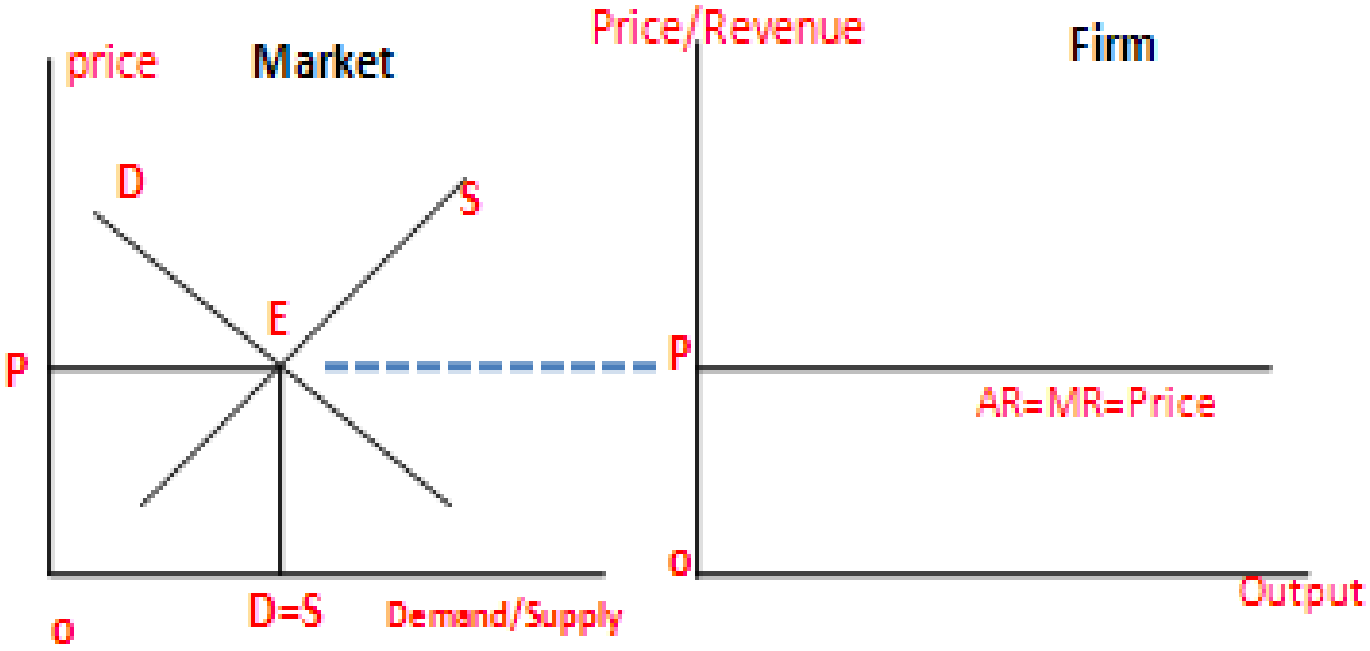
2. Homogeneous Products

Products offered for sale by all the sellers are homogenous/ identical in every respect . thus, the products of different sellers are perfect substitutes of one another

3. Uniform price

Price is uniform as the products of the different sellers in the market are homogenous. The price is determined forces of demand(buyer's bidding) and supply (sellers bidding) in the market and accepted by the all sellers are price takers in market. Hence firms under perfect competition are called price-takers.

Firms are price takers under perfect competition.



4. Free entry and exit

There is be no restriction on the entry and exit of both buyers to enter the market/industry.

5.Perfect knowledge about the market

Both buyers' and-sellers must have perfect knowledge about the conditions of the market such about ruling market price, the product. Hence, there is no selling cost incurred by the firms/sellers.

6. Perfect mobility

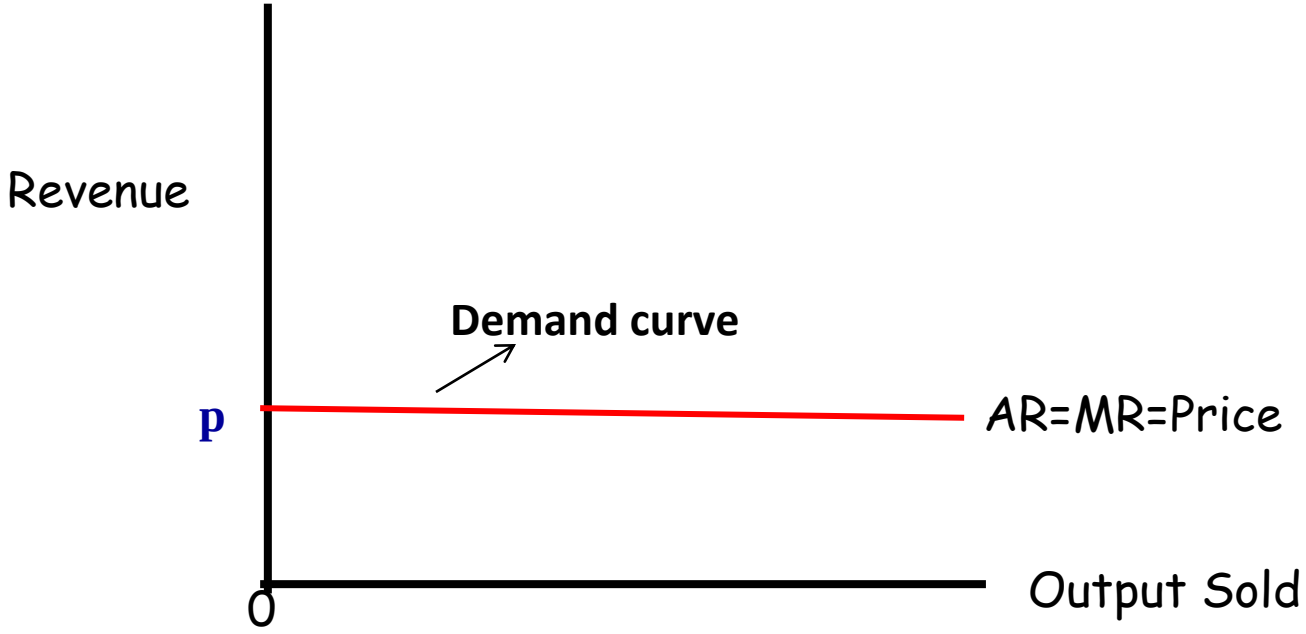
Various factors of production are perfectly mobile within the industry. Factors of production can freely move from one occupation to another and from one place to another, one firm to another. There is no barriers on their movement.

7. Absence of transport cost

Transport cost is zero. price of the product is not affected by the cost of transportation.

8. Perfectly elastic demand curve

Demand curve reflected by AR curve facing firm under perfect competition is perfectly elastic meaning that the firm can sell as much as it want at the ruling market price . Since price is uniform and given under perfect competition, both AR(price) and MR become equal. Thus, AR and MR curves coincide and become parallel to output axis.



Monopoly

Monopoly is a form of market structure where there is a single seller producing a commodity having no close substitute.

The word monopoly is derived from two Greek words-Mono and Poly. Mono means single and Poly means 'seller'. Thus monopoly means single seller.

MONOPOLY: FEATURES

1.One seller of a commodity

The characteristic feature of single seller eliminates the distinction between the firm and the industry. Single seller itself constitute the industry .

2. No close substitute

❑Product of the monopoly firm has no close substitutes .

3. Strong barriers to the entry into the industry

❑ Entry to the industry is restricted as the single firm imposes barriers to the entry of new and potential rivals in the industry.

4.The firm is price-maker

❑The firm is price-maker in the sense that the seller fixes the price of its product. The firm, thus, has independent price policy.

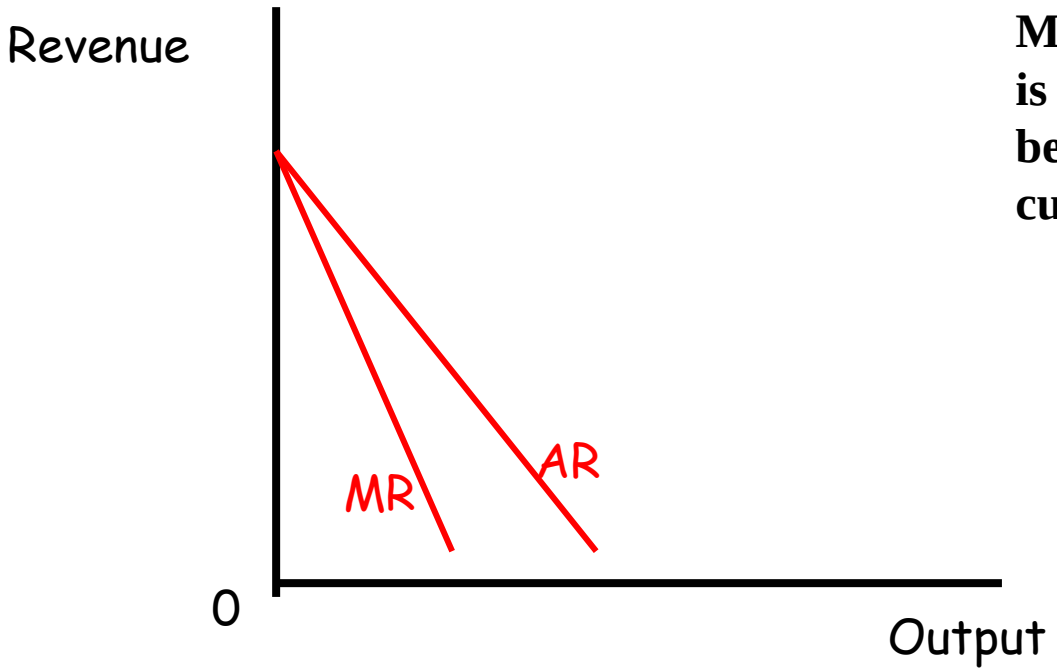
5.Price-discrimination may be possible

The firm may charge different prices from different set of customers for the same product or services.

(**Price discrimination is charging different prices for the same product or services from different consumers on the basis of personal, place and trade.)

6. Downward sloping inelastic demand curve of a monopoly firm

Demand curve of a firm reflected by its AR curve under monopoly is downward sloping meaning that the monopoly firm can sell more at a lower price and less at a higher price. The demand curve of the monopoly firm is highly inelastic. This is because the product does not have any close substitute.



MR curve of the monopoly firm is also downward sloping and lies bellow MR curve lies below AR curve.

Monopolistic Competition

Monopolistic competition is a market situation in which both monopoly and competitive elements are present.

The most distinguished features of monopolistic competition which makes it a blending of competition and monopoly is product differentiation.

(Product differentiation refers to the actively created differences in products with respect to brand, trademark, design, packing, colour, size, measurement, weight such that though the products are similar, they are not identical or in other words the products are different but closely related.**

Examples: Soaps lux, liril, cinthol, pears, hamam)

Bike (hero, honda, suzuki, bajaz and so on)

Monopolistic Competition: Features

1. Many sellers

There many sellers or firms in the industry such that individual firm is incompetent to affect the price of the industry.

2. Differentiated Products

Product differentiation is an important feature of monoploy. Products of different sellers are differentiated such that they are close substitute of each other with high degree of cross elasticises.

3. Relatively free entry and exit

New firm can enter the industry or any existing firm can leave the industry.

4. Each firm may have a tiny 'monopoly' because of the differentiation of their product

As the products are differentiated, they are unique in their own right. Each firm thus enjoys some degree of monopoly over its product.

Monopolistic Competition: Features(contd.)

5. Firm has some control over price

A firm under monopoly can influence the price of its product to some extent.

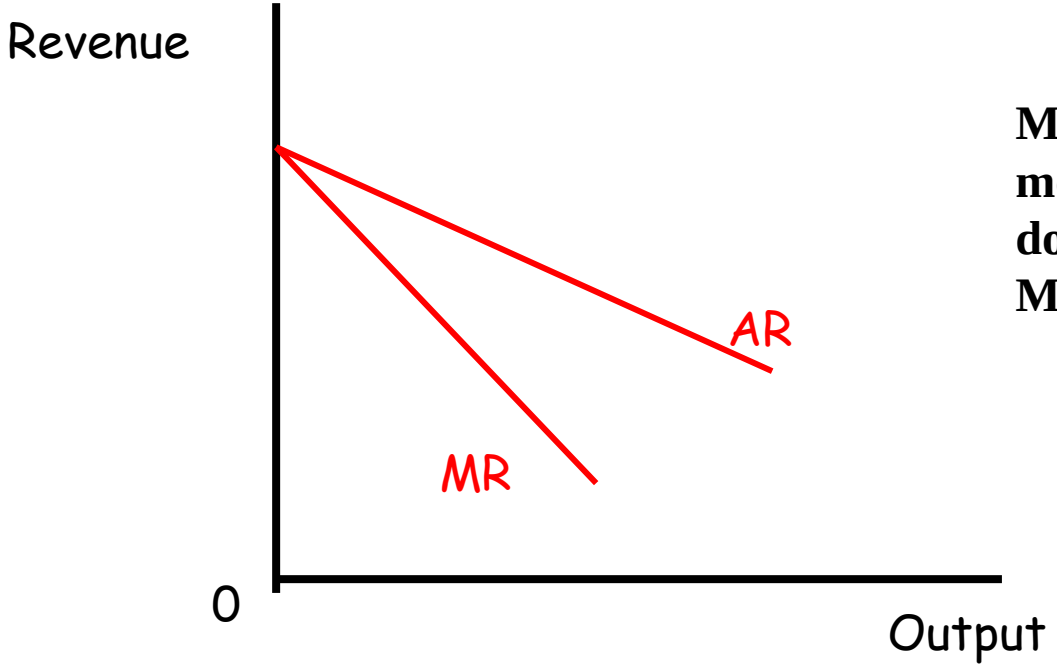
6. Presence of selling cost

The firms bears selling cost such as expenses on advertisement, publicity etc to create customer base and loyalty.

(* Selling costs are the expenses incurred on advertisement, publicity, salesmanship etc. Selling costs are incurred by the firms to make aware of the product, to attract new customers, to create customer base and loyalty).**

7. Downward sloping highly elastic demand curve of a firm under monopolistic competition

Demand curve of a firm reflected by its AR curve under monopolistic competition is downward sloping. The demand curve is highly elastic. This is because differentiated products under monopolistic competition has more close substitutes.



MR curve of a firm under monopolistic competition is also downward sloping and lies bellow MR curve lies below AR curve.

OLIGOPOLY

Oligopoly is a market situations in which there are few (more than two) sellers of a commodity such that actions of every seller has predictable effect on the other sellers/rivals.

Hence, oligopoly is also called competition amongst the few.

Oligopoly may be of two types can two forms:

- **pure oligopoly or oligopoly without product differentiation.**
- **differentiated oligopoly or oligopoly with product differentiation.**

Oligopoly : Features

1. Industry dominated by small number of large firms

2.High barriers to entry

3. Products may be highly differentiated (branding) or homogenous

4. Every seller has predictable impact on the other

5. Active presence of non-price competition such as selling and advertisement costs.

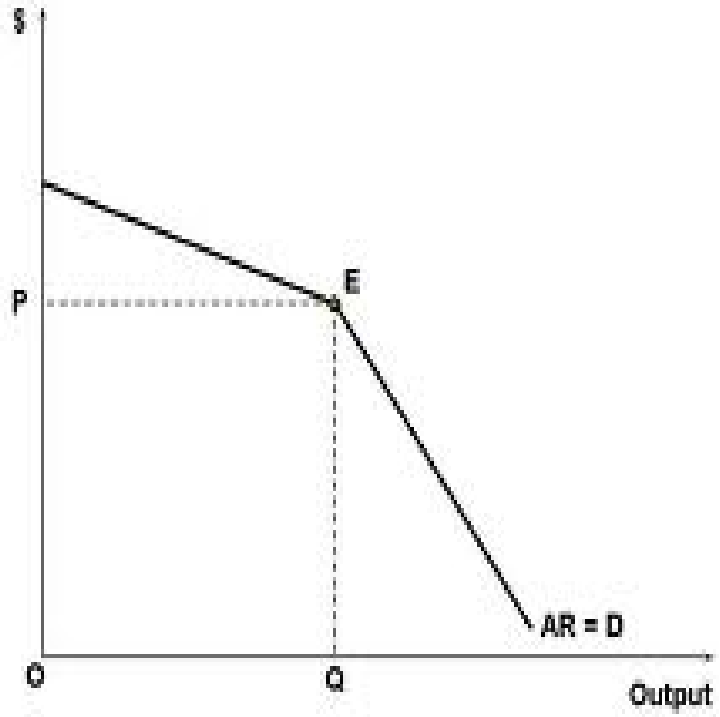
6. High degree of interdependence between firms and each firm take into account the reactions of others while making its own decisions

7. There may be price-rigidity (price stability) or price war within the market

DUOPOLY

Duopoly is a market situation in which there are two sellers of a commodity such that actions of each seller has predictable effect on the other seller/rival.

PRICE-RIGIDITY AND PAUL SWEEZY'S KINKED DEMAND CURVE



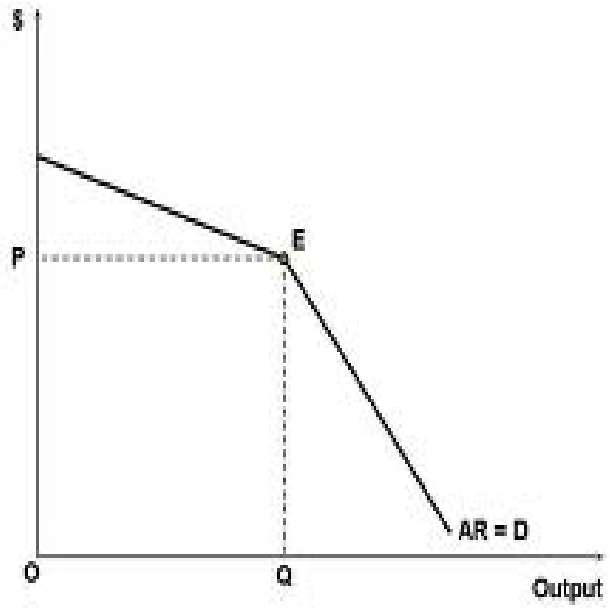
- Sweezy argued that an ordinary demand curve does not apply to oligopoly markets and promotes a kinked demand curve.

- Kink demand curve has a kink(E) at the prevailing level of price (P).

- the segment of the demand curve above the kink is highly elastic

- the segment of the demand curve bellow the kink is highly inelastic

Price-rigidity and Paul Sweezy's Kinked Demand curve contd.



- If the oligopoly firm reduce the price bellow the prevailing price (at kink), his rivals will follow him and accordingly lower their price. Very little increase in the sale can be obtained by a reduction in price by the oligopolist.

- If the oligopoly firm raises the price above the prevailing price (at kink), his rivals will not follow him. Large fall in sale is expected if the oligopolist by a increase in price.

- Hence the oligopolist has no incentive to raise its price or reduce it. Hence there is price rigidity/stable price in the market.

PROFIT , BREAK-EVEN POINT

AND

EQUILIBRIUM OF A FIRM

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PROFIT

Profit is defined as the gap between total revenue and total cost.

Profit(∏) =Total Revenue – Total Cost

When profit is negative, a firm is said to be suffering from loss.

BREAK -EVEN POINT

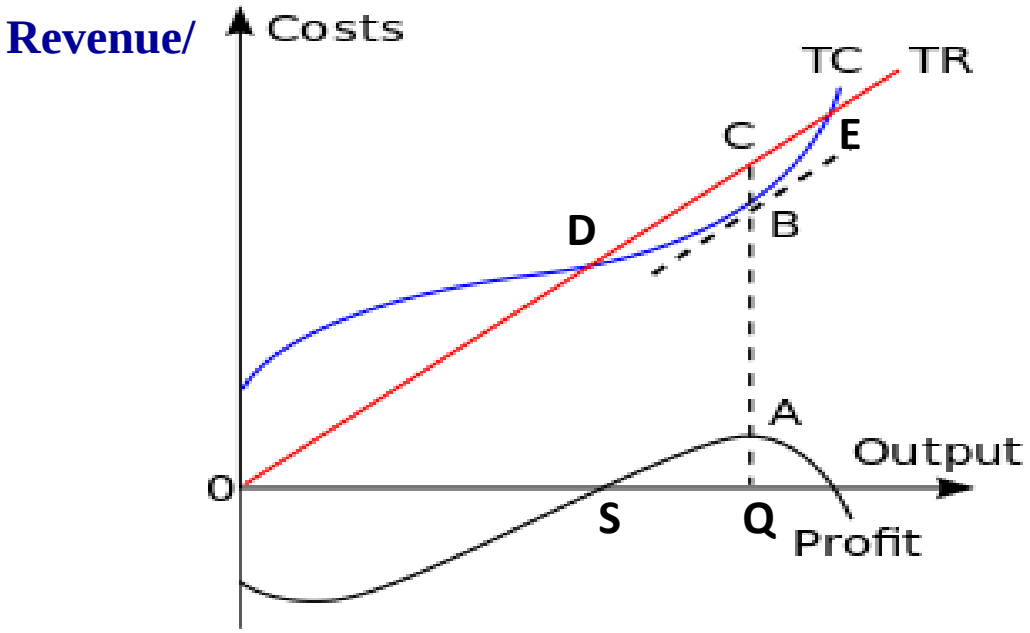
Break-even point indicates the level of output at which Total Revenue just equals Total Cost .

EQUILIBRIUM OF A FIRM

A firm is said to in equilibrium when it maximises its profit at given level of output.

Thus, at firm's equilibrium, profit(TR-TC) is maximum.

Break-Even Point and Firm's Equilibrium(Maximum Profit)



D(TR=TC) is lower break-even point at output S.

E (TR=TC) is upper break even point

To the left of lower break-even point D(output S), profit is negative.

When the firm increases its output beyond output S, the positive gap between TR and TC increase and profit accrue to the firm.

Break-Even Point and Firm's Equilibrium(Maximum Profit) (contd.)

The firm is in equilibrium (maximum profit =CB=AQ) at Q level of output after which the gap between TR and TC goes on narrowing. TR again becomes equal to TC at upper break even point E.

Upper break-even E point is not of much relevance as it lies beyond firm's profit maximizing level. Further, it lies beyond the firm's capacity.

The lower break even point D at output S is much significance as the firm would not plan to expand if it can not sell its output equal to at least S.

EQUILIBRIUM OF A FIRM

A firm is said to in equilibrium when it maximises its profit at given level of output.

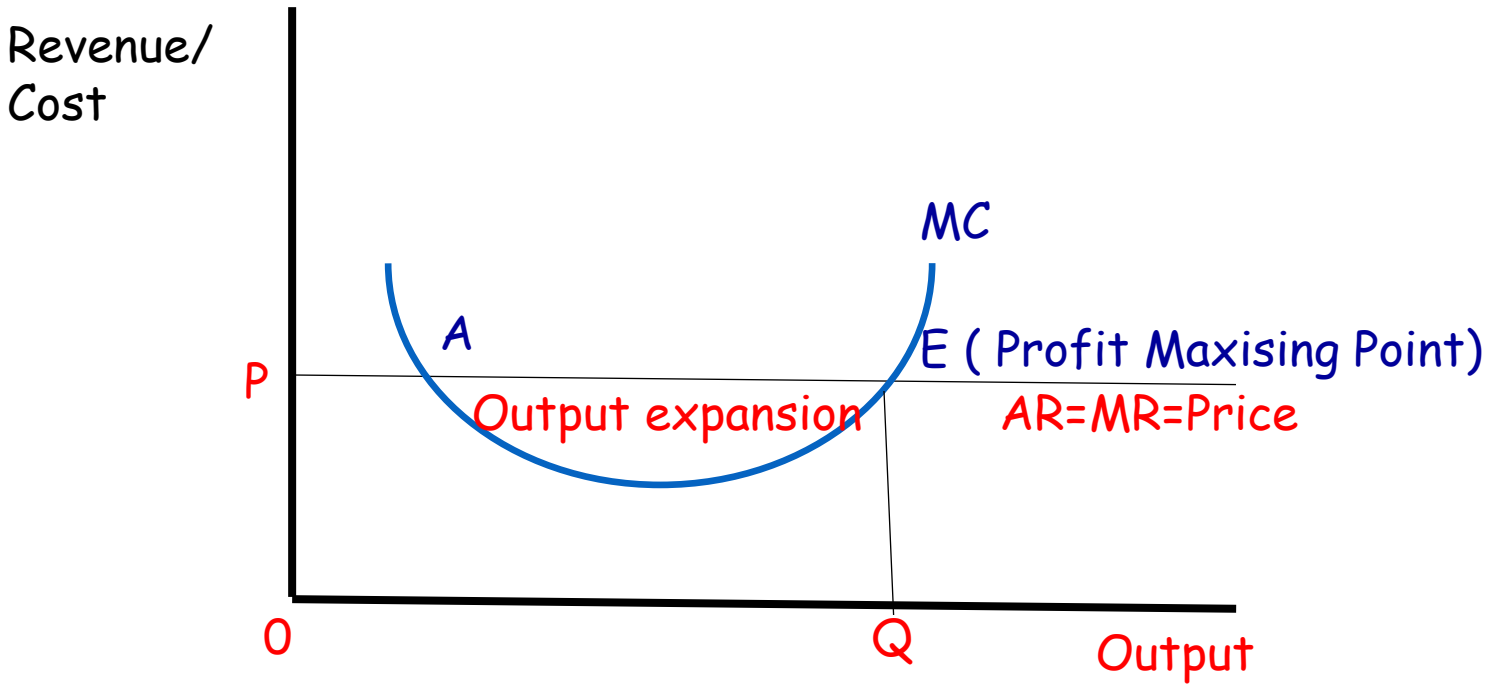
Thus, at firm's equilibrium, profit(TR-TC) is maximum.

CONDITIONS FOR PROFIT MAXIMISATION/FIRM'S EQUILBRIUM:

- **MR=MC (Necessary Condition)**

- **MC must be rising at the profit maximising/equilibrium point i.e MC curve must cut MR curve from bellow(Sufficient Condition)**

Equilibrium of a firm /Profit Maximisation by a firm (Marginal Approach contd.) : Graphical Illustration contd.



❖**Here at point A, MR=MC but MC cuts point A from above. Hence though necessary condition is satisfied , sufficient condition is not. Hence it is not profit maximising point.**

❖**Falling MC induces the firm to produce more and more till MC curve cuts MR curve from bellow.**

Equilibrium of a firm /Profit Maximisation : Graphical Illustration (contd.)

❖At point E , MR=MC and MC curve cuts MR curve from bellow. At E, both necessary and sufficient conditions are satisfied. Hence E is the point of equilibrium at which the firm maximises its profit.

Q is the equilibrium or profit maximising output. The firm will not produce beyond point E or output Q . Because after point E, MR < MC. As a result of which the firm will suffer loss.

EQUILIBRIUM OF A FIRM /PROFIT MAXIMISATION : MATHEMATICAL FORMULATION (CONTD.)

Conditions for Profit Maximisation:

Necessary Condition: $$\frac{d\prod}{dQ} = 0$$

Sufficient Condition: $$\frac{d^2(\prod)}{(dQ)^2} < 0$$

We have,

Profit($\prod$) = TR - TC

Let TR =R(Q) and TC=C(Q)

where Q is output.

Equilibrium of a firm /Profit Maximisation : Mathematical Formulation (contd.)

Conditions for Profit Maximisation:

At maximum profit,

Necessary Condition

$$\frac{d\prod}{dQ} = 0$$

$$\frac{d(TR - TC)}{dQ} = 0$$

$$\frac{d(R(Q) - C(Q))}{dQ} = 0$$

$$R'(Q) - C'(Q) = 0$$

Hence, MR=MC

Sufficient Condition

$$\frac{d^2(\prod)}{(dQ)^2} < 0$$

$$\frac{d}{dQ}\left(\frac{d\prod}{dQ}\right) < 0$$

$$\frac{d}{dQ}\left(\frac{d(R(Q) - C(Q))}{dQ}\right) < 0$$

$$R''(Q) - C''(Q) < 0$$

$$R''(Q) < C''(Q)$$

Hence, slope of MR< slope of MC